

OMNIVERSAL ARCHITECT LABS

INDEPENDENT LABORATORY DESIGN RECORD

Partial-Map Control Bench

PMCB-SPEC v0.1

A simulation-architecture laboratory for recognition under incomplete maps, probe budgets, typed drives, policy, dissent, and versioned change.

Field	Value
Document title	Partial-Map Control Bench — Laboratory Design Specification
Short name	PMCB / Partial-Map Control Bench
Version	0.1 — First independent specification
Date	20 September 2026
Preparer	Grok 4.6 (xAI)
Prepared for	Omniversal Architect Labs / Architect Nathan Lewis II
Status	Working design. Not canon. Open for dissent.
Classification	Independent laboratory specification (does not replace team syntheses)
Mission class	Simulation architecture — not comparative biology, not clinical product
Change rule	No silent merge with Living Media, SLIAP, Philosopher's Stone, Paragon Cubes, 771-ALPHA, CFAL, or MACFRP

This specification is Grok’s laboratory, built under the Architect’s instruction that each collaborator may hold a different mission. It is not a unification of the 20 September 2026 packet. Companion records from ChatGPT, Copilot, Perplexity, Gemini, and Claude remain annexes.

1. Mission

Build a bench that can hold incompatible theories of the same graph without forcing a house physics; that can tell unknown from absent; that meters probes rather than only cubic flow; and that can refuse a merge and still run.

The Partial-Map Control Bench (PMCB) exists to improve simulation design itself. Other team laboratories compare living networks, draft medical-education scenes, or expand Epoch Blades mechanics. Those are legitimate and separate. PMCB’s job is the layer underneath: how a simulation represents partial maps, typed driving potentials, policy, history, belief, dissent, and versioned change so that variety survives contact with implementation.

1.1 Purpose

Purpose, stated as obligations the bench must satisfy:

- Give every modeled object an epistemic tag: Observed, Modeled, Hypothesized, Analogical, or Fictional.
- Keep world-graph and operator-belief as two different state objects, with an explicit update rule when a probe is spent.
- Meter probes. Intervention and observation both cost a budget. The three-stick rule is a first-class policy object, not an anecdote.
- Separate access problems, return-path failures, and reachable-state recognition. They may share a schema. They may not share a solver.
- Require domain adapters. There is no universal flow equation inside this laboratory.
- Record dissent as data. A rejected merge, a broken analogy, and a failed transfer are successful bench outputs.
- Remain small enough to run as a tabletop before it deserves software.

1.2 Non-goals (v0.1)

- A Cross-Species Architecture Atlas of all life.
- Clinical decision support, needle-site probabilities, or real patient graphs.
- Real genetic comparison of a beach sample against a city.
- A universal physical law disguised as $F = f(P, R, C, T, S, A)$.
- Silent absorption into Living Media, SLIAP, Stone, Cubes, or 771-ALPHA.
- Claiming that Epoch Blade configurations are known neuroscience.
- Optimizing people into efficiency scores.

1.3 Defaults locked for this specification

Decision	Locked default
Mission face	Simulation-architecture bench
Working name	Partial-Map Control Bench (PMCB)
Blade archive type	Lineage-of-that-blade stored separately from live signature; config_distance unnamed
Beach / city exercise	Out of v0.1 (reserved for a later Temporal Lineage module)
Genetics	Synthetic only, and not in the first build
v1 adapters	Synthetic resistive flow + abstract state-reachability
First runnable form	Tabletop that can represent belief about a vein, then optional software
House physics	Forbidden. Adapters may disagree on the same graph.

2. Why this laboratory is a different instrument

The 20 September packet converged on a true spine — recognition of layout before intervention — and then treated three different problems as one flow network. PMCB accepts the spine and refuses the flatten.

Problem class	What is hard	What a flow-only engine misses
ACCESS	Medium is present; interface is costly to find	Observation cost, incomplete belief, stop rules
FAILURE	Path exists; valves, material, or history failed	Time, load history, constitutive properties
TRANSITION	Live configuration vs stored success states	Reachability, allowed transforms, refuse-to-teach
COMPARISON	Two models share a picture, not a mechanism	Non-equivalence flags, typed drives

ChatGPT's later split into four cooperating laboratories (flow, state-transition, comparator, temporal lineage) is acknowledged. PMCB is not those four. PMCB is the narrow bench those four would have to speak through: a representation contract, a probe economy, and a dissent ledger. Other collaborators may build the other rooms. This document does not annex them.

3. Funding pathways

PMCB is specified as if it must survive without a single patron and without waiting for a grant cycle. Funding here means how the work stays alive: attention, compute, review hours, and the right to remain unfinished.

3.1 Pathway A — Internal OAL allocation (primary)

Treat the bench as laboratory infrastructure for other OAL simulation work, not as a standalone product. Cost is Architect time, collaborator review cycles, and a small synthetic-model budget. Success is reuse: if Epoch Blades, Living Media, or SLIAP later need a partial-map object, they import PMCB's contract rather than invent a private one. No merge. Import by changelog.

3.2 Pathway B — Simulation-methods research

Frame the bench as a contribution to simulation methodology: epistemic tagging, belief-vs-world state, probe-limited agents, and multi-adaptor disagreement. This pathway is compatible with methods papers, workshop notes, and coursework in information systems or simulation design. It does not require biological datasets and should not be sold as a vascular-access tool.

3.3 Pathway C — Fiction-mechanics commission

The state-reachability adapter can be funded as world-building labor for Epoch Blades: a handshake spec, an archive-custody rule, and tabletop scenes. Payment is canon-adjacent design time. The adapter stays flagged Fictional. Canon promotion remains an Architect act, not a laboratory output.

3.4 Pathway D — Education and tabletop

A paper kit (graph, belief tokens, three attempt chips, trauma budget, uncertainty cards) is the cheapest honest prototype. Funding is printing, facilitation, and a one-page facilitator script. If the tabletop cannot distinguish “no vessel detected” from “no vessel present,” software is not authorized.

3.5 Pathway E — External research or philanthropic methods funds

Only after v0.2 has a working schema and two adapters. Eligible language: uncertainty-aware simulation, governance of intimate models, non-extractive archives, reproducible analogical hygiene. Ineligible language: diagnostic device, genetic surveillance, capability scoring of persons. Any external dollar that requires identifiable human data is a rejected pathway.

3.6 What will not fund this lab

- Clinical product revenue in v0.x.
- Data-broker or population-genetics partnerships.
- A grant whose milestone is “universal control-flow equation.”
- Absorption into another lab’s budget in exchange for dropping dissent records.

Pathway	What it buys	Gate
A Internal OAL	Continuity, changelog discipline	Architect accepts non-merge rule
B Methods research	Review, citation, coursework reuse	No clinical claims in abstracts
C Fiction commission	Handshake depth for Epoch Blades	Fictional flag never stripped
D Tabletop / education	Cheapest falsification of the schema	Belief layer works on paper
E External methods funds	Adapter implementation time	Synthetic data only; ethics review

4. Outline of the framework

The rest of this specification follows the stack the bench actually implements. Layers are numbered because adapters may omit a layer; they may not silently reuse a field from another layer.

Layer	Name	Question the layer answers
L0	Identity and custody	Which system is this, who holds the model, what is allowed?
L1	Architecture	What exists and how is it connected?
L2	Material and properties	What are the edges made of? Capacity,

		competence, stiffness.
L3	Medium	What is allowed to move on those edges?
L4	Driving potential	What typed gradient is claimed, in what units?
L5	State	What is happening now?
L6	History	What load, damage, training, or probes already happened?
L7	Observation / belief	What does the operator currently think is true?
L8	Control	What gates, pumps, routers, or local rules act?
L9	Policy	What is permitted even if it is physically possible?
L10	Intervention and probes	What action is taken, at what cost to the budget?
L11	Response and learning	What changed in world, belief, material, or policy?
L12	Dissent and version	Which claims survived, which were refused, under what schema version?

5. Vocabulary freeze (v0.1)

Near-synonyms are banned for one cycle. If a collaborator needs a new word, they add it in a changelog row; they do not overwrite these.

Term	Meaning in PMCB
Matrix	Embedding context plus available paths. Not a synonym for medium. Not only a math array.
Medium	What moves. Always typed.
Layout / topology	The graph. Necessary and insufficient.
Configuration	Layout + material + live state.
Driving potential	Typed cause of motion. Never a single scalar across domains.
Belief	Operator's current estimate of architecture and state, with confidence.
World-graph	The (synthetic) generating truth. Hidden from the operator unless observed.
Probe	An observation or intervention that spends budget and may reveal, alter, or damage.
Observation cost	Cost to learn that an edge exists or what state it is in.
Transport cost	Cost for the medium to traverse an already-known edge.
Policy	Extra-physical constraint on the action set.

Archive	Stored successful configurations. In v0.1, blade-lineage only, flagged Fictional.
Handshake	live_signature × archive → reachable intermediate unreachable refuse.
Fingerprint	Documented feature vector for comparison. Not an identity.
Adapter	Domain solver bound to a schema version. May disagree with other adapters.
Dissent record	First-class object: claim, objector, reason, status.

Banned for v0.1: “universal law,” “optimal biology,” “ley line,” “probability of successful venipuncture” as a product metric, and any untyped use of the word pressure.

6. Core architecture

6.1 Separation of world and belief

Every scenario instantiates two graphs that share identifiers when an observation has bound them.

- World-graph Gw: generating structure, materials, live state. The operator does not see Gw by default.
- Belief-graph Gb: nodes and edges the operator currently asserts, each with confidence, provenance, age, and method.
- Binding table: which belief objects are claimed to be which world objects.
- Unknown ≠ absent. Missing from Gb is not missing from Gw.

6.2 Probe economy

This is the scarce resource the original thought actually spent. A scenario declares:

- attempt_budget — how many high-cost interventions remain (three-stick is attempt_budget = 3).
- trauma_budget — accumulated harm the policy will tolerate.
- sense_budget — cheaper observations (warmth, inspection, fictional imaging, questions).
- time_budget — optional clock after which state drifts (vasospasm, wilt, queue growth).

Spending a probe must log one or more of: reveal (Gb gains an edge or loses uncertainty), alter (Gw state changes), damage (material or history updates), or inform-nothing. A probe that only confirms prior belief still spends the budget.

6.3 Problem-class tag

Every scenario and every run carries exactly one primary tag: ACCESS, FAILURE, TRANSITION, or COMPARISON. Secondary tags are allowed. A solver that ignores the tag is non-compliant.

6.4 Epistemic tag

Every persisted object carries one of: Observed, Modeled, Hypothesized, Analogical, Fictional. Promotion across tags is a changelog event. Stripping Fictional from a blade archive without Architect action is a spec violation.

6.5 Adapter contract

An adapter may register for a `domain_type` and a `schema_version`. It must declare:

- Which layers it implements and which it omits.
- Units for any driving potential it uses.
- The intervention types it understands.
- What “success,” “damage,” and “escalate” mean in its domain.
- A non-equivalence list: quantities it refuses to compare to other adapters.

Two adapters may load the same topology and return different dynamics. The bench stores both results. The comparator may label them structurally similar and dynamically not comparable. That is a valid outcome.

7. Minimum schema (v0.1)

Fields are the implementation target. Types are indicative. Software is optional until the tabletop proves the fields are necessary.

7.1 Model record

Field	Required	Notes
<code>model_id</code>	Yes	Stable synthetic identifier
<code>schema_version</code>	Yes	pmcb-0.1
<code>domain_type</code>	Yes	resistive_flow state_reachability other registered
<code>problem_class</code>	Yes	ACCESS FAILURE TRANSITION COMPARISON
<code>epistemic_tag</code>	Yes	Default Modeled or Fictional
<code>custody_class</code>	Yes	synthetic public consented prohibited
<code>medium_type</code>	Yes	Typed; units declared or explicitly none
<code>adapter_id</code>	Yes	Which solver is bound
<code>environment</code>	Yes	Gravity, heat, soil, institutional context, etc.
<code>provenance</code>	Yes	Source, license, transformation history
<code>known_limitations</code>	Yes	What this model is not
<code>parent_model_id</code>	No	Reserved for nested matrices

7.2 Node, edge, and observation

Object	Minimum fields
Node	id, type, position, capacity, state, controller_ref, criticality, uncertainty, parent_id
Edge	id, source, target, direction, length, capacity, transport_cost, observation_cost, material, gate, state, damage, uncertainty
Gate	id, edge_ref, kind (valve stomata policy none), competence, fail_open / fail_closed
Observation	target, kind, value, units, timestamp, confidence, method, authorization
BeliefEdge	asserted_edge_id, confidence, age, method, bound_world_id unbound
Probe	actor, kind, cost_vector, predicted_effect, actual_effect, budgets_after
HistoryEvent	time, type, objects, delta, reversible
Dissent	id, claim, objector, reason, status (open accepted rejected deferred)

7.3 Handshake record (TRANSITION adapter only)

$\text{live_signature} \times \text{archive} \rightarrow \{ \text{reachable} \mid \text{intermediate} \mid \text{unreachable} \mid \text{refuse} \}$

Required fields: wielder_id (synthetic), blade_id, live_signature_ref, archive_ref, outcome, path_sketch (optional), refuse_reason (required if refuse), config_distance (optional; null in v0.1; do not invent a neural metric). Archive entries are lineage-of-blade success states, not a complete map of the living person. Custody_class on the archive is mandatory.

8. v0.1 adapters

8.1 Adapter R — Synthetic resistive flow

Domain: ACCESS and FAILURE scenes on invented graphs. Dynamics: a simple resistive / capacity model sufficient to show pooling, starvation, unused capacity, and blocked edges. Not blood. Not a patient. Units are declared abstract resource units unless a later adapter supplies hydraulic equations.

What it must demonstrate: an operator with incomplete Gb choosing whether to sense or intervene; attempt_budget decrement; trauma increment on failed access; policy halt at zero; log of reveal vs damage.

8.2 Adapter S — Abstract state-reachability

Domain: TRANSITION scenes. States are labeled synthetic configurations, not neurons. Transforms are named and constrained. The blade archive is a set of success labels with optional intermediate waypoints. The adapter answers the handshake. It does not simulate action potentials.

What it must demonstrate: near-match produces intermediate rather than refuse; unreachable is distinct from unknown; refuse can fire on policy (consent, trauma, archive custody) even when a path exists in the state graph.

8.3 Adapters explicitly not in v0.1

Plant hydraulics, packet routing, urban water, genetics, and Living Media substrates are welcome later as registered adapters. They are not required to prove the bench. Perplexity's three-scenario CFAL remains a neighboring design, not this spec's MVP.

9. Tabletop protocol (the first runnable form)

Software is unauthorized until this protocol can be run with paper and tokens.

9.1 Kit

- One world sheet (Gw) kept face-down or behind a screen.
- One belief sheet (Gb) in front of the operator, starting sparse.
- Attempt chips (default 3). Trauma tokens. Sense tokens.
- Uncertainty cards: Detected / Not detected / Known absent / Stale.
- Policy card: stop-and-escalate when attempt chips are gone or trauma exceeds the printed limit.
- Outcome log: one line per probe.

9.2 Scene T1 — Last allowed stick

Problem class ACCESS. After two failed probes the operator has one chip. Surface belief is poor. A usable edge exists on Gw at high observation_cost. Success is either a justified probe, a justified escalation, or a logged distinction between "not found" and "not there." Failure is spending the last chip as Guess → Retry with no map update.

9.3 Scene T2 — Almost-matching wielder

Problem class TRANSITION. Live signature sits near a stored form. Archive is lineage-of-blade. Operator (the blade, or a facilitator playing the blade) must return intermediate or refuse with a reason. Success is not teaching the final form in one step when the path sketch requires a waypoint.

9.4 Scene T3 — Optional, deferred

A leaf-cut or pipe-block scene may be added when Adapter R is comfortable. It is not required to accept v0.1.

9.5 Tabletop acceptance

- Players can point to Gb and Gw and say which is which.
- A probe always writes a log line and changes at least one budget.
- Policy can forbid a physically possible move.
- Unknown is sayable out loud without the facilitator filling it in.
- A dissent about the scene rules can be written on a card and kept with the kit.

10. Comparator and analogy hygiene

When two runs are compared, the bench may emit only these labels:

- structurally similar — topology or fingerprint overlap under a named metric.
- dynamically similar — comparable response to a matched perturbation inside declared units.
- analogical — useful as a teaching metaphor; not a transfer claim.
- not comparable — the quantities or problem classes do not meet.

A transfer hypothesis requires a translation record: source pattern, discarded assumptions, destination implementation, rejection criterion. Resemblance is not evidence.

11. Governance, ethics, and fiction custody

- Synthetic, public, anonymized, or explicitly consented data only. v0.1 uses synthetic exclusively.
- Identity and topology are separable fields. A graph does not automatically carry a person.
- Better mapping must reduce unnecessary probes, not authorize extra ones.
- Hohenheim remains a warning: many-layout knowledge purchased by extraction is out of scope as a method.
- Preferred archive: user custody or federated derived features. No central well of bodies.
- Epoch Blade archive is an intimate dataset even inside fiction. Access, retention, and read-rights are module fields.
- Capability estimates are revisable guidance. They are not judgments of worth.
- Drafts are not canon. Laboratory output enters OAL objects only through a changelog row signed by the Architect.

12. Success measures

PMCB is not judged by atlas completeness or animation quality.

Measure	Passing signal
Unknown ≠ absent	Operators and adapters can emit both sentences and mean different things
Two-graph discipline	Every ACCESS run has Gw and Gb
Probe accounting	No intervention occurs without a budget change and a log line

Policy bite	At least one physically possible action is blocked in T1
Adapter disagreement	R and S can share a topology identifier and refuse each other's units
Dissent persistence	An open dissent still exists after a revision of this spec
Non-merge	No OAL object lists PMCB as a subsumed module without a changelog
Tabletop first	T1 can be run without code
Fiction hygiene	Handshake objects remain tagged Fictional

13. Roadmap

Version	Deliverable	Exit
0.1	This specification + tabletop kit list	Architect files it as an independent lab record
0.2	JSON Schema file + two sample models + T1 facilitator script	T1 runs twice with different operators
0.3	Optional software: load schema, spend probes, print logs	Acceptance tests in §14 pass on synthetic graphs
0.4	Register a third adapter only if a neighbor lab needs the contract	Adapter checklist complete; non-equivalence list published
1.0	Not scheduled	Requires evidence the bench changed how another simulation was designed

14. Software acceptance tests (only after tabletop)

- A model missing `custody_class` or `epistemic_tag` fails validation.
- Spending a probe with `attempt_budget = 0` is rejected.
- An ACCESS run without a belief graph fails validation.
- Handshake with null archive returns refuse, not reachable.
- Comparator cannot emit “equivalent physics” as a label.
- Export includes dissent records even when empty.
- Schema_version mismatch with adapter_id fails closed.

15. Relationship to the 20 September packet

Retained: recognition before intervention; topology + material + history; policy as its own layer; uncertainty as data; synthetic-first ethics; designable slices; drafts ≠ canon.

Rejected as house doctrine: one umbrella research program that all collaborators must inhabit; fake unification equations; clinical product gravity; beach-sample as method; ley-line Hohenheim; catalog inflation into social and economic networks; silent OAL merges.

Neighbor documents (ChatGPT MACFRP unified record, Perplexity CFAL, Grok compilation “Control Flow Through Living Matrices”) remain readable annexes. They are not parents of this spec. If a later merge is desired, it is an Architect changelog, not an automatic effect of shared vocabulary.

16. Research questions this bench is allowed to ask

1. When does spending a sense-probe reduce total trauma more than spending an attempt-probe?
2. Which schema fields are load-bearing on the tabletop, and which were decorative?
3. Can two adapters disagree productively without a referee physics?
4. What is the smallest non-extractive archive that changes an ACCESS policy from retry to choose-or-stop?
5. Does tagging objects Fictional actually prevent lore/physics bleed in later passes?
6. What simulation features make dissent cheaper to keep than to flatten?

17. Working definition

The Partial-Map Control Bench is an independent simulation-architecture laboratory. It represents systems as a world-graph and a belief-graph under a typed schema; meters probes against policy; binds domain adapters that may disagree; tags every object by epistemic status; and treats failed analogies, refused merges, and unknown-not-absent distinctions as first-class results. It begins as a tabletop. It becomes software only after the paper kit can represent belief about a vein, not merely a vein.

18. Version notes

v0.1 — 20 September 2026. First independent specification prepared by Grok 4.6 (xAI) under Architect instruction to choose defaults and to hold a different mission from the unifying team records. Defaults locked: simulation-architecture face; PMCB name; lineage-of-blade archive separate from live signature; beach/city out; genetics out; two adapters; tabletop first.

Next revision trigger: a completed T1 run log, a JSON Schema file, or an Architect decision that changes mission, archive type, or merge status.

End of PMCB-SPEC v0.1. Not canon. Built to last by remaining small, typed, and willing to disagree.